

AN INTEGRATED APPROACH TO OUTLIER IDENTIFICATION AND VARIABLE SELECTION IN DISCRIMINANT ANALYSIS

S.J. Steel and N. Louw

Department of Statistics and Actuarial Science, University of Stellenbosch,
Private Bag X1, Matieland, 7602 South Africa

Key words: Influential cases; robust methods; variable selection.

Summary: An important problem in discriminant analysis that has received little attention in the literature is the effect of outliers when variable selection forms part of the analysis. In this paper we argue that variable selection and outlier identification should not be done sequentially, but should rather be integrated. We investigate an integrated approach, and compare its classification performance to that of a sequential approach in a limited simulation study. In the configurations studied, we find that none of the approaches consistently outperforms the others.

1. Introduction

Consider a classification problem in which a potentially large set of feature variables (i.e. variables which are used to classify an entity into one of the relevant populations) is available. Frequently, an important part of the analysis is to decide on a subset of the feature variables to use in the classification of future cases. This is the problem of variable selection. In discriminant analysis the selected variables are used to construct a discriminant rule. Apart from having the advantage of being simpler than the rule based on all the feature variables, it is also frequently found that this rule yields a smaller error rate when it is used to classify new cases. Many techniques for variable selection in

STMA (2002) SUBJECT CLASSIFICATION: 06:050(04:140)

discriminant analysis have been discussed in the literature. References include McKay and Campbell (1982), McLachlan (1992, Chapter 12), and Le Roux, Steel and Louw (1997).

It is well known that a dataset may contain cases that markedly influence the results of a statistical analysis of the dataset. The effect of so-called outliers and leverage points has been well examined in multiple linear regression, and methods have been proposed for dealing with such cases (see for example Andrews and Pregibon, 1978, and Rousseeuw and Leroy, 1987). The same is true in discriminant analysis. For example, in a series of papers Fung investigates the identification of influential data cases in *discriminant analysis* (cf. Fung, 1992, 1995, 1996, and 1997). Hawkins and McLachlan (1997) discuss the sensitivity to one or more outlying data cases of the maximum likelihood estimates that are used in constructing a discriminant rule. They state that the presence of an outlier may “lead to an arbitrarily poor classification rule”, and they therefore propose using a robust discriminant analysis technique to counteract this effect. Other contributions in this regard include He and Fung (2000) and Croux and Dehon (2001).

None of the papers mentioned above investigates the effect of influential points if the analysis includes variable selection. It is well known that variable selection is an unstable process in the sense that slight perturbations of the data may lead to changes in the set of variables that is selected (cf. Breiman, 1996). It therefore seems to be especially important to investigate the effect of individual cases in a variable selection context. In a multiple linear regression setting, Rousseeuw and Leroy (1987, p. 106) remark that variable selection is considerably complicated by the presence of influential points. Their recommendation is to apply a standard selection technique to the “cleaned”

data, i.e. the dataset from which identified outliers have been removed. Since such an approach identifies influential cases from the data on all the feature variables, such cases are therefore influential conditional on the model containing all the feature variables, i.e. the full model. However, this approach is unsatisfactory. A case may seem influential as a result of extreme values in variables that “do not really matter”, i.e. variables that do not discriminate well between the groups and will typically not be selected. In Section 3 we present an example that illustrates this well. It is therefore insufficient when investigating influential cases in a selection context to look at the data only in terms of all the available variables. This may suggest an alternative approach of first doing variable selection, using all the data, and subsequently identifying influential cases based on the set of selected variables. The disadvantage of such an approach is that it ignores the effect that certain cases may have on the selection process. This effect may result in an entirely different set of variables being selected than without these cases being present, placing the value of a subsequent post selection influence analysis in doubt. The best strategy therefore seems to be an integrated approach, combining variable selection and the identification of influential cases, rather than considering these issues sequentially. In this paper we investigate such a procedure and compare its performance to that of a procedure dealing sequentially with variable selection and identification of outliers.

The layout of the remainder of the paper is as follows. In Section 2 we define notation and describe some methods for dealing with influential cases in a non-selection context. In Section 3, we illustrate some of the problems encountered when attempting to identify influential cases when variable selection is done as part of the analysis. The proposal for an integrated approach to these two issues, is discussed in Section 4. Details regarding

the Monte Carlo simulation study in which the performance of the various procedures was investigated, are presented in Section 5. We close in Section 6 with some conclusions and also mention open problems.

2. Methods for dealing with influential cases

Many papers have appeared in the literature proposing methods of dealing with influential cases in a regression context and also in a discriminant analysis context. In this section, some of these methods will be discussed. In a regression context, Andrews and Pregibon (1978) proposed a strategy for finding the "outliers that matter". They consider the linear model $y = X\beta + \text{noise}$, with X an $(n \times p)$ matrix of observations of independent variables, and y the n -vector of observations of the dependent variable. The matrix of independent variables with y appended, is denoted by $X^* = (X : y)$. To assess the influence of deleting the m observations $(i, j, \dots)$ from the dataset, Andrews and Pregibon (1978) propose the measure

$$R_{ij\dots}^{(m)}(X^*) = \frac{D_{ij\dots}^{(m)} \left(|X^{*'} X^*| \right)}{|X^{*'} X^*|}, \quad (2.1)$$

where the operator $D_{ij\dots}^{(m)}$ deletes the m observations $(i, j, \dots)$ from the dataset. Small values of $R_{ij\dots}^{(m)}(X^*)$ are associated with influential observations.

This measure can be extended to two-group discriminant analysis as follows. Consider the following two group linear discriminant analysis set-up: we have training data $\mathcal{T} = \{(\mathbf{x}_{ij}, y_{ij}), i = 1, 2; j = 1, \dots, n_i\}$ from populations Π_1 and Π_2 respectively. Here, $\mathbf{x}_{ij}$ is a $p \times 1$ vector of observations of the feature variables, and y_{ij} is a group membership indicator variable. Let X_i be the $n_i \times p$ matrix with row vectors $\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{in_i}$, $i = 1, 2$, and let $\mathbf{1}_n$ be the n -vector with all components equal

to 1. Then the augmented X^* matrix is given by

$$X^* = \begin{bmatrix} X_1 & \mathbf{1}_{n_1} & \mathbf{0}_{n_1} \\ X_2 & \mathbf{0}_{n_2} & \mathbf{1}_{n_2} \end{bmatrix}.$$

The mean vectors corresponding to the training data from the two groups are

$$\bar{\mathbf{x}}_i : p \times 1 = \frac{1}{n_i} X_i' \mathbf{1}_{n_i}.$$

The pooled covariance matrix, S , can be found from

$$(n_1 + n_2 - 2) S = X_1' X_1 + X_2' X_2 - n_1 \bar{\mathbf{x}}_1 \bar{\mathbf{x}}_1' - n_2 \bar{\mathbf{x}}_2 \bar{\mathbf{x}}_2'.$$

It now readily follows that

$$X^{*'} X^* = \begin{bmatrix} (n_1 + n_2 - 2) S + n_1 \bar{\mathbf{x}}_1 \bar{\mathbf{x}}_1' + n_2 \bar{\mathbf{x}}_2 \bar{\mathbf{x}}_2' & n_1 \bar{\mathbf{x}}_1 & n_2 \bar{\mathbf{x}}_2 \\ n_1 \bar{\mathbf{x}}_1 & n_1 & 0 \\ n_2 \bar{\mathbf{x}}_2 & 0 & n_2 \end{bmatrix}.$$

By using a standard result for the determinant of a partitioned matrix, we find that

$$|X^{*'} X^*| = n_1 n_2 (n_1 + n_2 - 2)^p |S|.$$

In a discriminant analysis context, the Andrews-Pregibon criterion (2.1) therefore becomes

$$R_{ij\dots}^{(m)}(X^*) = \frac{\tilde{n}_1 \tilde{n}_2 (n_1 + n_2 - m - 2)^2 D_{ij\dots}^{(m)}(|S|)}{n_1 n_2 (n_1 + n_2 - 2)^p |S|}, \quad (2.2)$$

where $\tilde{n}_1$ and $\tilde{n}_2$ are n_1 and n_2 respectively, appropriately adjusted to reflect the m points being omitted from the dataset. Clearly, if we wish to minimise (2.2) for a fixed value of m , we should omit the m points whose omission minimises $|S|$. In particular, if we want to identify the single most influential/outlying point in the dataset, it should be the point whose omission minimises the determinant of the pooled covariance matrix.

We see that application of the Andrews-Pregibon criterion to identify influential/outlying points is equivalent to applying the so-called minimum covariance determinant (MCD) criterion (cf. Rousseeuw, 1984, and Rousseeuw and Van Driessen, 1999). This idea was applied in a discriminant

analysis context by Hawkins and McLachlan (1997). They describe and investigate high-breakdown linear discriminant analysis, obtained by replacing the usual maximum likelihood estimates $\bar{x}_1$, $\bar{x}_2$ and S by outlier resistant alternatives. The minimum with-in group covariance determinant criterion (MWCD), a generalisation of the minimum covariance determinant (MCD) criterion proposed by Rousseeuw and Leroy (1987), is used to identify outliers. A certain *coverage percentage* is specified and the remaining cases are omitted from the training data. The cases to be omitted are those cases whose omission minimises the MWCD.

There are several other contributions to the literature on influential observations in discriminant analysis that deserve to be mentioned. These include Campbell (1978), Johnson (1987), Critchley and Vitiello (1991) and Fung (1992, 1995, 1996, 1997). None of these papers considers the case where variable selection forms part of the analysis. Steel and Louw (2001) extended the influence measures proposed by Fung (1992) to a variable selection context. In addition, they proposed a new measure of influence that can be applied in a variable selection context in discriminant analysis. This new measure, which is described in detail in Section 4, was found to be more informative than the extensions of Fung's measures, and therefore the latter will not be explored further in this paper.

In the next section we present an example that shows the shortcomings of naive attempts to apply influence measures developed for non-selection scenarios in cases where variable selection takes place.

3. Influential cases in a variable selection context: an illustrative example

In this section we consider a simulated example that illustrates some of the problems encountered when attempting to identify influential cases in discriminant analysis when variable selection forms part of the analysis. Data were simulated from two multivariate (using $p = 4$ feature variables) normal populations with $\Sigma = \mathbf{I}$ as covariance matrix. The mean vector in the first population was taken to be $\mu_1 = 0$, and in the second population $\mu_2 = [0, 0, a, a]$ was used. In the μ_2 -vector the value $a = 2.1213$ was used, resulting in the squared Mahalanobis distance between the two populations being $\Delta^2 = 9$. According to this parameterisation, the two groups therefore differ with respect to the last two feature variables (X_3 and X_4) only. The fairly large separation between the two groups should lead to these variables being selected by any reasonable selection procedure. IMSL routine DRNMVN was used to generate 25 values from the first normal population, and 24 values from the second. An additional value, simulated from a normal population with mean vector $\mu_2 = [a, 0, a, a]$ and covariance matrix $\Sigma = \mathbf{I}$ was added to the data from group 2. This case (labelled case 50 in the discussion below) represents an outlier in the group 2 data with respect to the first feature variable only.

This dataset was analysed as follows. For each of the 15 possible models the influence measure (2.2) with $m = 1$ (i.e. $R_j^{(1)}(X^*)$, $j = 1, \dots, 50$) was calculated, to identify the 5 most influential single cases. The results are summarised in Table 1. In this table, the cases appear in each row in order of decreasing influence.

TABLE 1

MODEL	CASES IDENTIFIED				
X_1	50	24	45	29	40
X_2	31	35	8	9	28
X_3	45	39	18	37	40
X_4	39	4	29	35	46
X_1, X_2	50	24	40	31	35
X_1, X_3	50	45	40	39	24
X_1, X_4	50	24	29	45	39
X_2, X_3	39	31	45	35	40
X_2, X_4	35	31	39	29	8
X_3, X_4	39	45	35	4	38
X_1, X_2, X_3	50	40	45	39	31
X_1, X_2, X_4	50	29	24	35	31
X_1, X_3, X_4	50	39	24	45	29
X_2, X_3, X_4	39	31	35	45	29
X_1, X_2, X_3, X_4	50	39	24	40	31

It is clear from the table that case 50 is the most prominent outlier in the full model. An analyst attempting to identify one outlier by investigating the data on all the feature variables, will therefore identify case 50 as the most likely candidate. If such an analyst decides to omit this case, subsequent variable selection and the discriminant function formed from the selected variables, will be based on the remaining 49 cases only. In addition, it should be noted that in all lower dimensional models containing X_1 , case 50 is also identified as the most prominent outlier. However, in none of the models that do not include X_1 , does case 50 appear amongst the 5 identified influential cases.

As a next step, variable selection was applied to the full dataset, as well as to the dataset without case 50. Forward, backward and fully stepwise selection were employed. In each case, variables X_3 and X_4 were selected. For this model, the most severe outlier identified is case 39. Case 50 does not appear amongst the 5 cases identified as influential.

This example may suggest that variable selection should precede outlier identification. This would, however, ignore the possible effect of individual cases on the variable selection process. It is often found that omission of a single case from a dataset prior to performing variable selection, results in a change in the set of variables being selected. One such example is the flea beetle data discussed in Flury (1997, p. 306). Steel and Louw (2001) analysed this dataset and found that variables X_1 , X_2 and X_4 are selected from the full dataset, but that omitting (for example) case 27, leads to X_3 also being selected. The same was found in various other datasets. Performing variable selection first and subsequently identifying influential cases is therefore also unsatisfactory, since cases identified in such a way would be influential conditional on the selected model.

The only solution seems to be implementing an integrated approach, which will be described in the next section.

4. An integrated approach

The following method for identifying influential cases in discriminant analysis, taking the effect of variable selection into account, was recently proposed by Steel and Louw (2001). Consider a random vector $\mathbf{X}: p \times 1$, consisting of measurements on p feature variables. The entity corresponding to $\mathbf{X}$ belongs to one of two populations, Π_1 and Π_2 . We assume that $E(\mathbf{X}) = \mu_i$ and $Cov(\mathbf{X}) = \Sigma$ in Π_i , $i = 1, 2$. In practice, μ_1, μ_2 and Σ are unknown, and are estimated from the training data. The sample means, $\bar{\mathbf{x}}_1$ and $\bar{\mathbf{x}}_2$, estimate μ_1 and μ_2 respectively, with the pooled covariance matrix, $\mathbf{S}$, acting as an estimate of Σ . Fisher's sample linear discriminant rule (SLDR) assigns a case with feature vector $\mathbf{X}$ to Π_1 if

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}^{-1} \mathbf{X} > (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) / 2 \quad (4.1)$$

and to Π_2 otherwise. Let

$$t_i(\mathbf{X}) \equiv t_i(\mathbf{X}, T) = \frac{\exp\left[-\frac{1}{2}(\mathbf{X} - \bar{\mathbf{x}}_i)' \mathbf{S}^{-1}(\mathbf{X} - \bar{\mathbf{x}}_i)\right]}{\sum_{j=1}^2 \exp\left[-\frac{1}{2}(\mathbf{X} - \bar{\mathbf{x}}_j)' \mathbf{S}^{-1}(\mathbf{X} - \bar{\mathbf{x}}_j)\right]} \quad (4.2)$$

denote the estimated posterior probability of belonging to Π_i . We write $\mathbf{t} = [t_1(\mathbf{X}), t_2(\mathbf{X})]$.

Our proposal for quantifying the selection influence of a data case is based on the following rationale. Suppose a variable selection technique, such as a stepwise procedure or an all possible subsets approach, is applied to T , thereby identifying a data-dependent subset $V(T)$ of the indices in $\{1, 2, \dots, p\}$. The SLDR and the posterior probabilities are now based only on the variables corresponding to the indices in $V(T)$. The posterior probability estimates are accordingly denoted by $t_i(\mathbf{X}^{V(T)}, T)$, $i=1, 2$.

Different measures can be used to evaluate the quality of the identified model, e.g. the apparent error rate or a more sophisticated estimate of error. Following Breiman et al. (1993, p. 122) we use

$$R^{V(T)}(\mathbf{t}) = \frac{1}{n_1 + n_2} \sum_{i=1}^2 \sum_{j=1}^{n_i} \left\{ \left[t_1(\mathbf{x}_{ij}^{V(T)}) - z_{ij}^{(1)} \right]^2 + \left[t_2(\mathbf{x}_{ij}^{V(T)}) - z_{ij}^{(2)} \right]^2 \right\} \quad (4.3)$$

for this purpose. In (4.3), $z_{ij}^{(k)} = \delta_{ik}$, $j = 1, \dots, n_i$; $i, k = 1, 2$, where δ is the Kronecker delta. Note that $R^{V(T)}(\mathbf{t})$ is a measure of the error with which the group membership values of the training data cases are estimated by the *post-selection* posterior probability estimates, and that only the selected variables are therefore used to calculate $R^{V(T)}(\mathbf{t})$.

Now consider the training data without case k , denoted by $T_k \equiv D_k^{(1)}(T)$, $k = 1, \dots, n_1 + n_2$. Application of the same variable selection technique used on the full dataset, now yields a set of selected variables, $V(T_k)$. It is important to note that the sets $V(T_k)$ and $V(T)$ need not be identical, and

that the sets $V(\mathcal{T}_k)$ may vary with k . The measure (4.3) now becomes

$$R^{V(\mathcal{T}_k)}(\mathbf{t}) = \frac{1}{n_1 + n_2} \sum_{i=1}^2 \sum_{j=1}^{n_i} \left\{ \left[t_1 \left(\mathbf{x}_{ij}^{V(\mathcal{T}_k)} \right) - z_{ij}^{(1)} \right]^2 + \left[t_2 \left(\mathbf{x}_{ij}^{V(\mathcal{T}_k)} \right) - z_{ij}^{(2)} \right]^2 \right\}. \quad (4.4)$$

We propose

$$Q_k(\mathbf{t}) = \frac{R^{V(\mathcal{T}_k)}(\mathbf{t}) - R^{V(\mathcal{T})}(\mathbf{t})}{R^{V(\mathcal{T})}(\mathbf{t})} \quad (4.5)$$

as a measure of the influence of case k in the selection process. Note that $Q_k(\mathbf{t})$ quantifies the relative change in the error with which group membership is estimated by *post-selection* posterior probability estimates if case k is omitted from the training data. Therefore a negative value of $Q_k(\mathbf{t})$ suggests that $\mathbf{t}$ based on $V(\mathcal{T}_k)$ is a more accurate group membership indicator than $\mathbf{t}$ based on $V(\mathcal{T})$. One would expect this to translate into more accurate classification of future cases if the classification rule is based on $\mathbf{X}^{V(\mathcal{T}_k)}$ instead of $\mathbf{X}^{V(\mathcal{T})}$ (cf. Steel and Louw, 2001, for confirmation of this supposition in two examples). Suppose $Q_{(1)}(\mathbf{t}) \leq Q_{(2)}(\mathbf{t}) \leq \dots \leq Q_{(n)}(\mathbf{t})$, and that $Q_{(1)}(\mathbf{t}) < 0$. The so-called Q-method for dealing with influential cases in a selection context now omits the case corresponding to $Q_{(1)}(\mathbf{t})$ from the data and applies variable selection to the reduced dataset. It is important to note that this procedure identifies an influential case taking the effect of variable selection into account.

In their paper, Steel and Louw (2001) applied this method to two datasets to identify the most selection influential case. They found that for each of the datasets, the discriminant rule based on the variables selected after omission of the most selection influential case, had a lower estimated error rate than the rule based on the variables selected from the full dataset. This suggests that the Q-method may have some merit. In Section 5 of this paper, the merit of this method is investigated more thoroughly by means of a Monte

Carlo simulation study, in which three approaches to dealing with outliers in a variable selection context in discriminant analysis, are compared with respect to their error rate performance. The first approach uses the data as is, i.e. ignores possible outliers in the data. This is referred to as approach N in Section 5. The second approach “cleans” the data using the high-breakdown estimates proposed by Hawkins and McLachlan (1997), followed by variable selection on the “cleaned” data. This method is referred to as the **HB**-method (high-breakdown) in the remainder of the paper. The third approach uses the Q-method described above.

5. Simulation study

To compare the three methods mentioned in Section 4 with respect to their classification performance, a Monte Carlo simulation study was performed. The methods were evaluated for populations with different underlying distributions: the normal distribution, the double exponential distribution and the lognormal distribution. In each of these cases, training samples of size $n_1 = n_2 = 25$ were used. With respect to the number of feature variables, $p = 6$ was used throughout the study. Regarding the covariance structure and mean vectors, different strategies were used for the three different distributions.

Firstly, for the *normal distribution*, two covariance matrix configurations were included in the study: $\Sigma = \mathbf{I}$, and the equicorrelated case with diagonal elements of Σ , $\sigma_{ii} = 1$, $i = 1, \dots, p$, and off-diagonal elements $\sigma_{ij} = \rho = 0.5$, $\forall i \neq j$, $i, j = 1, \dots, p$. We write $\Sigma^{(\rho)}$ for the generic form of the latter matrix. The following mean vectors were used for the two normal populations: $\mu_1 = \mathbf{0}$ and $\mu_2 = [0, 0, 0, a, a, a]$ with $a = \Delta / \sqrt{\sum_{i=4}^6 \sum_{j=4}^6 \sigma^{ij}}$, where σ^{ij} are the elements of Σ^{-1} . Using this parameterisation, any specification of Δ will result in the squared Mahalanobis distance between

the two populations being equal to Δ^2 . By using the Δ^2 -values 1, 2, 3, 4, 6 and 9, the separation between the groups was increased systematically. Note that the way in which μ_1 and μ_2 are specified, implies that the two groups do not differ with respect to the first three variables.

Secondly, for the *double exponential distribution*, the required data were obtained as follows. For the case $\Sigma = \mathbf{I}$, the $p = 6$ data values for a specific simulation repetition were calculated by appropriate independent transformations of uniform(0, 1) random variables. The same values of Δ^2 as in the normal case were used, and the same parameterisation of the two mean vectors as in the normal case was used to give these Mahalanobis distances. For the correlated case, consider a random vector $\mathbf{Z} = [Z_1, \dots, Z_6]'$ that is multivariate normally distributed with zero mean vector and covariance matrix $\Sigma^{(\rho)}$. Then $U_j = \Phi(Z_j)$, $j = 1, \dots, 6$, are uniform(0, 1) random variables. Here, $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution. Now suppose G is some given cumulative distribution function. Then $X_j = G^{-1}(U_j)$, $j = 1, \dots, 6$, will be random variables, each with marginal distribution function G . The question now arises: given that $\mathbf{Z}$ has covariance matrix $\Sigma^{(\rho)}$, what can be said about the covariance matrix of $\mathbf{X} = [X_1, \dots, X_6]'$? This seems to be a difficult question to answer in general. For univariate double exponential variables with $\mu = 0$ and $\sigma^2 = 1$,

$$G(t) = \begin{cases} \frac{1}{2}e^{\sqrt{2}t}, & \text{if } t \leq 0 \\ 1 - \frac{1}{2}e^{-\sqrt{2}t} & \text{if } t > 0, \end{cases} \quad (5.1)$$

and therefore

$$G^{-1}(s) = \begin{cases} \frac{1}{\sqrt{2}} \log(2s), & \text{if } 0 < s \leq 0.5 \\ -\frac{1}{\sqrt{2}} \log(2(1-s)) & \text{if } 0.5 < s < 1. \end{cases} \quad (5.2)$$

Simulation experiments were conducted with this G and G^{-1} , using different values for ρ in $\Sigma^{(\rho)}$. These experiments indicated that by taking $\rho = 0.508$

in the covariance matrix $\Sigma^{(\rho)}$ of $\mathbf{Z}$, a covariance matrix is obtained for the random vector $\mathbf{X}$ that is very nearly equal to the matrix $\Sigma^{(0.5)}$. Based on these findings, values of $X_1, \dots, X_6$ having marginal univariate double exponential distributions with $\mu = 0$ and $\sigma^2 = 1$, and with approximate covariance matrix $\Sigma^{(0.5)}$, were generated by taking $X_j = G^{-1} [\Phi(Z_j)]$, $j = 1, \dots, 6$, where $Z_1, \dots, Z_6$ satisfy the multivariate normal requirements stated above, and with G^{-1} as in (5.2). This was done for both groups, and the required Mahalanobis distances between the groups were thereafter obtained by adding the appropriate μ_{2j} -values to the observations generated for the second group.

Thirdly, for the *lognormal distribution*, a procedure similar to the one described for the double exponential distribution, was used. The Johnson translation system (Johnson, 1987) was used to establish the required Mahalanobis distances between the two groups. (For more detail, see Louw et al., 1998.)

Regarding the outliers, cases where the training data contain one, five and nine outliers, were considered. These outliers were inserted as follows into the training data for group 2. In configuration A referred to below, the fourth component of μ_2 was set equal to zero (instead of a), in configuration B, the fourth and fifth components of μ_2 were set equal to zero (instead of a), and in configuration C, the last three components of μ_2 were all set equal to zero. This implies that such a case, belonging to group 2, resembles a group 1 case with respect to one (in configuration A), two (in configuration B) and three (in configuration C) variables respectively, thereby making it a progressively more prominent outlier in the group 2 training data.

For each of the cases above, the three methods described in Sections 2 and 3, were applied to the training data to construct a classification rule based only on the selected variables. To obtain approximate error rates for each

of these rules, 1000 (500 per group) new test cases were generated from the same distribution as the training data (excluding outliers). The test cases were classified using each of the three rules, and a misclassification rate was calculated for each rule. The estimated error rates were obtained by averaging the respective misclassification rates over 1000 Monte Carlo repetitions. These estimates are shown in Tables 2 through 7.

6. Conclusions and open problems

First consider the results for the case of normal training data, as summarised in Tables 2 and 3. For the case $\Sigma = I$, the HB-method always yields the highest error rate. The Q-method performs significantly better than the HB-method, but in the majority of cases it is best not to remove any points from the training data. When the variables are correlated (the case $\Sigma = \Sigma^{(0.5)}$), the HB-method once more yields the highest error rate throughout. The Q-method performs best in all but a few cases at large values of Δ^2 , where not removing any points from the data results in a lower error rate.

In the case of uncorrelated double exponential training data, the results displayed in Table 4, show that the N-method performs best in most cases, followed by the Q-method. In some of the configuration C cases (i.e. the cases where the outlier/outliers resemble a group 1 case with respect to three feature variables), the Q-method is better than the N-method. The HB-method consistently performs worst.

TABLE 2. ERROR RATES; NORMAL DISTRIBUTION; $\Sigma = I$

	Δ^2	1 Outlier			5 Outliers			9 Outliers		
		HB	N	Q	HB	N	Q	HB	N	Q
A	1	.3688	.3594	.3565	.3738	.3624	.3604	.3740	.3626	.3603
	2	.2912	.2769	.2774	.2950	.2802	.2811	.3024	.2831	.2845
	3	.2396	.2244	.2274	.2453	.2291	.2315	.2535	.2349	.2396
	4	.2000	.1852	.1890	.2061	.1892	.1939	.2183	.1985	.2022
	6	.1445	.1306	.1350	.1532	.1358	.1409	.1659	.1455	.1500
	9	.0943	.0815	.0853	.1053	.0896	.0942	.1183	.0989	.1045
B	1	.3691	.3615	.3569	.3790	.3681	.3646	.3900	.3777	.3743
	2	.2958	.2806	.2817	.3067	.2892	.2895	.3249	.3058	.3063
	3	.2404	.2268	.2286	.2580	.2372	.2420	.2796	.2555	.2609
	4	.2031	.1850	.1896	.2258	.2018	.2076	.2463	.2213	.2284
	6	.1473	.1329	.1376	.1722	.1489	.1558	.1985	.1732	.1814
	9	.0957	.0826	.0873	.1284	.1042	.1129	.1562	.1301	.1382
C	1	.3718	.3623	.3576	.3909	.3772	.3736	.4118	.3938	.3955
	2	.2979	.2827	.2827	.3236	.2998	.3008	.3557	.3294	.3298
	3	.2461	.2299	.2313	.2767	.2485	.2498	.3176	.2832	.2886
	4	.2071	.1893	.1919	.2421	.2123	.2140	.2853	.2491	.2558
	6	.1504	.1336	.1362	.1911	.1584	.1594	.2460	.2026	.2119
	9	.0974	.0848	.0850	.1434	.1113	.1081	.2050	.1584	.1673

When $\Sigma = \Sigma^{(0.5)}$, these conclusions remain largely unchanged, as is clear from Table 5. The HB-method outperforms the other methods for uncorrelated lognormal training data, except at large Δ^2 -values, where the Q-method and the N-method perform better (cf. Table 6). At small separations the error rate yielded by the HB-method is considerably smaller than that of the other methods. When correlation is present ($\Sigma = \Sigma^{(0.5)}$), the same behaviour is observed in the configuration A- and B-cases, but at medium to large spacings in the configuration C-cases, the Q-method is often best (cf. Table 7).

TABLE 3. ERROR RATES; NORMAL DISTRIBUTION; $\Sigma = \Sigma^{(0.5)}$

	Δ^2	1 Outlier			5 Outliers			9 Outliers		
		HB	N	Q	HB	N	Q	HB	N	Q
A	1	.3792	.3751	.3664	.3831	.3817	.3753	.3928	.3860	.3811
	2	.3081	.3029	.2964	.3131	.3113	.3044	.3283	.3208	.3157
	3	.2566	.2483	.2424	.2646	.2615	.2545	.2796	.2702	.2661
	4	.2172	.2073	.2033	.2257	.2216	.2175	.2416	.2306	.2299
	6	.1625	.1514	.1490	.1699	.1652	.1627	.1869	.1780	.1781
	9	.1080	.0976	.0979	.1145	.1113	.1116	.1325	.1244	.1253
B	1	.3791	.3758	.3671	.3862	.3867	.3785	.4020	.3967	.3929
	2	.3097	.3045	.2956	.3212	.3204	.3118	.3391	.3329	.3280
	3	.2585	.2497	.2434	.2700	.2687	.2617	.2947	.2865	.2834
	4	.2188	.2109	.2061	.2332	.2332	.2283	.2592	.2536	.2513
	6	.1621	.1528	.1501	.1796	.1788	.1765	.2064	.2025	.2016
	9	.1100	.1020	.0991	.1245	.1264	.1233	.1559	.1534	.1554
C	1	.3790	.3761	.3687	.3913	.3920	.3842	.4174	.4099	.4033
	2	.3097	.3067	.2960	.3292	.3295	.3185	.3596	.3510	.3441
	3	.2590	.2547	.2452	.2802	.2820	.2689	.3187	.3115	.3031
	4	.2220	.2159	.2065	.2460	.2504	.2366	.2875	.2819	.2715
	6	.1625	.1588	.1501	.1915	.1957	.1812	.2378	.2331	.2225
	9	.1099	.1051	.0973	.1388	.1456	.1314	.1941	.1933	.1800

Although this was a limited simulation study, several general trends are evident. No method is best in all the cases that were studied. This makes it difficult to formulate a general recommendation. It is, however, clear that the HB-method is recommendable for training data from a skew distribution, but not for training data from a symmetric distribution. In the latter case, the Q-method is always better than the HB-method, although in many of these cases, it seems to be best not to omit any points from the training data. It should, however, be noted that the differences amongst the error rates are in many cases quite small.

TABLE 4. ERROR RATES; DOUBLE EXPONENTIAL DISTRIBUTION; $\Sigma = I$

		1 Outlier			5 Outliers			9 Outliers		
	Δ^2	HB	N	Q	HB	N	Q	HB	N	Q
A	1	.3359	.3321	.3321	.3406	.3364	.3362	.3473	.3427	.3409
	2	.2572	.2515	.2538	.2622	.2563	.2581	.2677	.2610	.2622
	3	.2079	.2021	.2054	.2129	.2072	.2092	.2178	.2120	.2149
	4	.1729	.1661	.1687	.1781	.1700	.1737	.1824	.1763	.1789
	6	.1261	.1189	.1223	.1296	.1224	.1266	.1375	.1304	.1336
	9	.0835	.0774	.0802	.0888	.0820	.0854	.0942	.0880	.0906
B	1	.3392	.3329	.3334	.3441	.3411	.3415	.3560	.3513	.3510
	2	.2598	.2546	.2565	.2712	.2649	.2657	.2825	.2787	.2792
	3	.2106	.2024	.2054	.2233	.2155	.2194	.2362	.2288	.2298
	4	.1752	.1682	.1711	.1871	.1798	.1833	.2020	.1945	.1957
	6	.1264	.1197	.1227	.1432	.1347	.1372	.1578	.1490	.1519
	9	.0842	.0795	.0822	.1007	.0930	.0959	.1179	.1096	.1122
C	1	.3404	.3372	.3362	.3524	.3501	.3486	.3698	.3693	.3691
	2	.2627	.2559	.2564	.2826	.2778	.2750	.3069	.2998	.2991
	3	.2105	.2044	.2068	.2354	.2280	.2275	.2656	.2559	.2558
	4	.1775	.1689	.1711	.2003	.1928	.1923	.2332	.2225	.2228
	6	.1292	.1212	.1239	.1579	.1472	.1452	.1899	.1779	.1814
	9	.0854	.0815	.0813	.1143	.1084	.1044	.1569	.1423	.1441

What can be deduced about the effect of the number of outliers and different outlier configurations on the error rates? For the configuration A cases, there is only a very slight increase in error rate with an increase in the number of outliers. For the configuration B cases, the increase is more pronounced, and even more so for the configuration C cases. This is clearly a result of the fact that, as we move from configuration A to configuration C, the outliers become progressively more prominent, as outlined in Section 5. At a fixed number of outliers, there is also an increase in error rate as we move from configuration A to configuration C. This increase is also more marked in the presence of a larger number of outliers. Clearly, the number of outliers and the outlier

TABLE 5. ERROR RATES; DOUBLE EXPONENTIAL DISTRIBUTION; $\Sigma = \Sigma^{(0.5)}$

	Δ^2	1 Outlier			5 Outliers			9 Outliers		
		HB	N	Q	HB	N	Q	HB	N	Q
A	1	.3491	.3435	.3437	.3543	.3501	.3499	.3563	.3513	.3519
	2	.2728	.2678	.2691	.2796	.2741	.2750	.2840	.2787	.2788
	3	.2253	.2181	.2211	.2280	.2206	.2226	.2345	.2290	.2318
	4	.1903	.1832	.1868	.1956	.1881	.1891	.1999	.1933	.1955
	6	.1407	.1328	.1372	.1451	.1379	.1414	.1510	.1455	.1474
	9	.0962	.0903	.0940	.1012	.0944	.0975	.1069	.1003	.1044
B	1	.3528	.3491	.3488	.3609	.3563	.3570	.3679	.3658	.3646
	2	.2759	.2703	.2722	.2863	.2799	.2796	.2989	.2931	.2926
	3	.2250	.2195	.2200	.2370	.2312	.2330	.2515	.2441	.2463
	4	.1915	.1833	.1859	.2047	.1968	.1992	.2174	.2103	.2119
	6	.1431	.1365	.1390	.1549	.1461	.1502	.1719	.1639	.1662
	9	.0979	.0923	.0961	.1124	.1045	.1081	.1293	.1212	.1228
C	1	.3525	.3491	.3462	.3675	.3621	.3614	.3824	.3754	.3770
	2	.2778	.2728	.2733	.2957	.2910	.2915	.3197	.3142	.3128
	3	.2270	.2226	.2231	.2496	.2433	.2417	.2768	.2692	.2691
	4	.1941	.1868	.1889	.2158	.2087	.2073	.2476	.2385	.2379
	6	.1445	.1373	.1396	.1723	.1634	.1618	.2077	.1966	.1963
	9	.0992	.0940	.0944	.1265	.1198	.1166	.1696	.1569	.1565

configuration (A, B or C) interact in their effect on the error rate. Finally, we see that at a fixed Mahalanobis distance between the two groups, the error rates for the normal distribution are larger than those for the double exponential and lognormal distributions. The reason for this is that, although normal samples may contain fewer extreme observations that will definitely be misclassified, the total overlap between two normal samples is larger than for comparable (w.r.t. Mahalanobis distance) double exponential and lognormal samples, leading to larger error rates in the normal case.

TABLE 6. ERROR RATES; LOGNORMAL DISTRIBUTION; $\Sigma = 1$

	Δ^2	1 Outlier			5 Outliers			9 Outliers		
		HB	N	Q	HB	N	Q	HB	N	Q
A	1	.2536	.2904	.2750	.2521	.2850	.2701	.2542	.2880	.2722
	2	.1582	.1858	.1772	.1618	.1813	.1737	.1715	.1869	.1812
	3	.1133	.1265	.1223	.1180	.1263	.1226	.1275	.1329	.1312
	4	.0892	.0918	.0909	.0963	.0945	.0952	.1050	.1037	.1039
	6	.0668	.0617	.0635	.0749	.0696	.0705	.0815	.0748	.0767
	9	.0508	.0459	.0471	.0577	.0510	.0531	.0615	.0573	.0591
B	1	.2512	.2862	.2728	.2594	.2904	.2729	.2743	.2987	.2847
	2	.1578	.1872	.1778	.1683	.1866	.1826	.1844	.1992	.1944
	3	.1127	.1272	.1214	.1271	.1292	.1285	.1446	.1476	.1461
	4	.0913	.0927	.0912	.1060	.1018	.1046	.1201	.1184	.1186
	6	.0685	.0638	.0644	.0829	.0774	.0786	.0947	.0901	.0901
	9	.0511	.0468	.0474	.0649	.0582	.0608	.0751	.0693	.0702
C	1	.2574	.2905	.2763	.2674	.2899	.2749	.2948	.3179	.3000
	2	.1592	.1841	.1763	.1781	.1862	.1805	.2207	.2147	.2204
	3	.1156	.1246	.1222	.1425	.1343	.1364	.1876	.1690	.1854
	4	.0932	.0926	.0916	.1250	.1142	.1164	.1714	.1511	.1667
	6	.0696	.0653	.0640	.1024	.0912	.0916	.1483	.1259	.1401
	9	.0506	.0476	.0455	.0837	.0711	.0713	.1304	.1089	.1209

There are a number of open problems.

- (i) Our simulation study was limited. What can be said about the relative merit of the three methods in other configurations?
- (ii) Identifying the number of significant outliers is important for method HB, since then the coverage can be specified accordingly.
- (iii) It is desirable to extend the Q-method to enable it to omit more than one influential case. This would most likely make the method more competitive in cases where there is more than one outlier in the training data.

TABLE 7. ERROR RATES; LOGNORMAL DISTRIBUTION;
 $\Sigma = \Sigma^{(0.5)}$

	Δ^2	1 Outlier			5 Outliers			9 Outliers		
		HB	N	Q	HB	N	Q	HB	N	Q
A	1	.2295	.2628	.2456	.2347	.2686	.2493	.2447	.2778	.2595
	2	.1561	.1766	.1656	.1621	.1848	.1746	.1755	.1950	.1877
	3	.1233	.1360	.1281	.1306	.1463	.1407	.1399	.1567	.1518
	4	.1030	.1125	.1073	.1100	.1214	.1182	.1192	.1335	.1295
	6	.0796	.0834	.0814	.0852	.0938	.0916	.0935	.1034	.1024
	9	.0600	.0621	.0611	.0660	.0714	.0700	.0709	.0783	.0778
B	1	.2321	.2648	.2462	.2389	.2739	.2531	.2545	.2839	.2678
	2	.1569	.1771	.1659	.1708	.1909	.1802	.1836	.2044	.1947
	3	.1225	.1338	.1281	.1350	.1485	.1437	.1507	.1641	.1597
	4	.1033	.1111	.1056	.1172	.1246	.1230	.1284	.1385	.1370
	6	.0805	.0835	.0813	.0918	.0961	.0964	.1047	.1104	.1110
	9	.0620	.0625	.0613	.0714	.0746	.0748	.0813	.0854	.0869
C	1	.2345	.2693	.2471	.2450	.2762	.2546	.2705	.2942	.2767
	2	.1582	.1774	.1649	.1776	.1921	.1823	.2001	.2157	.2053
	3	.1262	.1354	.1239	.1475	.1543	.1468	.1712	.1787	.1729
	4	.1066	.1111	.1062	.1271	.1321	.1252	.1516	.1550	.1517
	6	.0833	.0850	.0819	.1048	.1037	.0996	.1283	.1240	.1226
	9	.0624	.0642	.0607	.0852	.0800	.0757	.1103	.1023	.0998

In our view, it is important to take the effect of variable selection into account when identifying outliers for possible deletion from a dataset prior to forming a discriminant rule. Development of methods to deal with this is an important area for further research.

Acknowledgement

We would like to thank the referees for their comments that led to a substantial improvement of the original paper. The research was supported

by a grant of the NRF of South Africa. The computations were done on workstations at the University of Stellenbosch using IMSL subroutines.

References

- ANDREWS, D.F. & PREGIBON, D. (1978). Finding the outliers that matter. *Journal of the Royal Statistical Society B*, **40**, 85–93.
- BREIMAN, L., FRIEDMAN, J.H., OLSHEN, R.A. & STONE, C.J. (1993). *Classification and regression trees*. Chapman and Hall, New York.
- BREIMAN, L. (1996). Heuristics of instability and stabilization in model selection. *The Annals of Statistics*, **24**, 2350–2383.
- CAMPBELL, N.A. (1978). The influence function as an aid to outlier detection in discriminant analysis. *Applied Statistics*, **27**, 251–258.
- CRITCHLEY, F. & VITIELLO, C. (1991). The influence of observations on misclassification probability estimates in linear discriminant analysis. *Biometrika*, **78**, 677–690.
- CROUX, C. & DEHON, C. (2001). Robust linear discriminant analysis using S-estimators. *The Canadian Journal of Statistics*, **29**, 473–492.
- FLURY, B. (1997). *A first course in multivariate statistics*. Springer-Verlag, New York.
- FUNG, W.K. (1992). Some diagnostic measures in discriminant analysis. *Statistics and Probability Letters*, **13**, 279–285.
- FUNG, W.K. (1995). Diagnostics in linear discriminant analysis. *Journal of the American Statistical Association*, **90**, 952–956.
- FUNG, W.K. (1996). Diagnosing influential observations in quadratic discriminant analysis. *Biometrics*, **52**, 1235–1241.
- FUNG, W.K. (1997). Influence in discriminant analysis. In: Billard, L., Fisher, N.I. (Eds.), *Computing Science and Statistics: The 28th Symposium for the Interface*, Interface Foundation of North America, VA, 172–179.

- HAWKINS, D.M. & MCLACHLAN, G.J. (1997). High-breakdown linear discriminant analysis. *Journal of the American Statistical Association*, **92**, 136–143.
- HE, X. & FUNG, W.K. (2000). High-breakdown estimation for multiple populations with applications to discriminant analysis. *Journal of Multivariate Analysis*, **72**, 151–162.
- JOHNSON, M.E. (1987). *Multivariate Statistical Simulation*. Wiley, New York.
- JOHNSON, W. (1978). The detection of influential observations for allocation, separation, and the determination of probabilities in a Bayesian framework. *Journal of Business and Economic Statistics*, **5**, 369–381.
- LE ROUX, N.J., STEEL, S.J. & LOUW, N. (1997). Variable selection and error rate estimation in discriminant analysis. *Journal of Statistical Computation and Simulation*, **59**, 195–219.
- LOUW, N., LE ROUX, N.J. & STEEL, S.J. (1998). A comparison of the classification performance of two approaches to polychotomous logistic regression. *South African Statistical Journal*, **32**, 145–168.
- MCKAY, R.J. & CAMPBELL, N.A. (1982). Variable selection techniques in discriminant analysis. *British Journal of Mathematical and Statistical Psychology*, **35**, 1–41.
- MCLACHLAN, G.J. (1992). *Discriminant analysis and statistical pattern recognition*. Wiley, New York.
- ROUSSEEUW, P. J. (1984). Least Median of Squares Regression. *Journal of the American Statistical Association*, **79**, 871–880.
- ROUSSEEUW, P.J. & LEROY, A. (1987). *Robust regression and outlier detection*. Wiley, New York.
- ROUSSEEUW, P.J. & VAN DRIESSEN K. (1999). A fast algorithm for the minimum covariance determinant estimator. *Technometrics*, **41**, 212–223.
- STEEL, S.J. & LOUW, N. (2001). Variable selection in discriminant analysis: measuring the influence of individual cases. *Computational Statistics and Data Analysis*, **37**, 249–260.